Companion A to the Blank-Slate Audit: Objections Answered, Tolerances Made Explicit, and Sources Graded by Kind

New computations, an objection audit, and a concept ledger for "The Universe and Consciousness"

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11 November 2026

Abstract This document is a companion to *The Universe and Consciousness: A Blank-Slate Audit* (43 pages, 13 sections, 18 figures). It does three things that the main paper had no room for. First, it answers the objections a reader will actually raise: eighteen of them, each answered with either a number I computed in this pass or an explicit statement that I cannot answer it and why. Second, it makes my own choices measurable. Where the main paper said "the coincidence is not sharp," this companion computes the whole curve of how sharp it is as a function of the tolerance I am free to grant, and where the main paper quoted a pooled H_0, this companion shows what every defensible pooling estimator returns on the same four numbers, including a fully marginalised Bayesian model, and how far the pooled value moves when each experiment is deleted in turn. Third, it grades the source pool by *kind of object* — dataset, peer-reviewed primary, peer-reviewed reanalysis, preprint, curated reference work, blog — with a worked verification of one case (the PBR theorem) that also produced a small lesson about withdrawn comments. Five new figures, all generated from the computations themselves, are embedded. Everything here is reproducible with one command, named in the appendix.

Provenance note, stated up front because it affects what is and is not re-verified here. Between the second and third pass the sandbox that held my working tree was reset: the scripts, logs and the 18 embedded PNGs of the main paper were lost from the workspace (they survive in the uploaded evidence kit and inside the published PDF, which is the reason I uploaded a kit at all). The analysis in this companion was therefore rebuilt from the raw data files — re-downloaded from the same GitHub release — in a new script, `deepen.py`, and cross-checked against the numbers printed in the main PDF. Section 3.4 reports what that independent reimplementation caught, in both directions.

What this pass was for

The brief I was given was to deepen the audit, diversify and grade the sources, deepen the concepts, and leave the reader no opening for a question. That last instruction is impossible in the literal reading and worth taking seriously in the intended one: I cannot prevent questions, but I can *pre-empt* them, which means anticipating the objection, answering it with the cheapest computation that settles it, and writing down the answers I do not have in the same font as the answers I do. So this companion is organised as follows. Section 2 states the epistemic status of every kind of claim in it. Section 3 reports four new computations. Section 4 grades the source pool by kind, with the PBR verification as a worked example. Section 5 is the concept ledger: fifteen terms that carry weight in the main paper, each defined exactly, then said again in plain words, then fenced off from the misuse I have seen it suffer. Section 6 is the objection audit. Section 7 lists what I refuse to conclude and what would change my mind, updated with the numbers below. Appendix A is reproduction.

Two conventions from the main paper carry over unchanged, because they are the point of the exercise. Every claim is tagged: **[FACT]** (established, checked against a primary source), **[FACT, CITED, NOT FETCHED]** (established, but relying on my reading of a source I could not open in this session), **[MY CALCULATION]** (arithmetic I ran here, with the log line findable by searching its own digits), **[MY MEASUREMENT]** (a value obtained by fitting data I downloaded), and **[SPECULATION]** (inference, mine or somebody else's, that outruns the above). And every quantitative claim in this companion has a figure, five of them, with the caption generated from the same JSON that the plot was drawn from.

Evidence tiers, and what each one forbids

A tier system is only worth having if it constrains the sentence that uses it, so I will state the constraint rather than the definition. **[FACT]** may be used as a premise in an argument. **[FACT, CITED, NOT FETCHED]** may be used to report a position but not to refute one — I will not tell a philosopher that they are wrong about their own paper if I have only read about it. **[MY CALCULATION]** may be used to draw a conclusion only if the same paragraph states the model that the calculation is internal to; my 10^{122} entropy comparison is a calculation inside general relativity plus quantum field theory in a fixed background, and it buys no argument about whether that is the right framework. **[MY MEASUREMENT]** comes with the additional requirement that I report how many parameters I fitted and what the goodness-of-fit null looks like, which is Section 3.1. **[SPECULATION]** may not appear in a sentence containing a number unless the number is the thing being speculated about.

Four new computations

Is my supernova fit "too good"? A goodness-of-fit null, simulated

The main paper reports a two-parameter fit to the 40 binned Pantheon distance moduli with $\chi^2 = 39.30$ on 38 degrees of freedom, and a reader is entitled to ask two opposite questions at once: is that too high (is the model wrong) and is that too low (is my covariance overestimated, or did I

sneak in freedom)? A table of $\chi^2(38)$ answers the first question and an answer of my own making answers the second.

I generated 4,000 synthetic 40-bin Hubble diagrams by taking my own best-fit model and adding correlated noise drawn from the published 40×40 covariance, then refitting each one with exactly the code that produced the number in the text. The null distribution of the refit statistic has median 37.38 and a 90% band of $[25.18, 53.04]$; my observed 39.30 sits at the 58.6th percentile, with $P(\chi^2_{\rm null} \geq \chi^2_{\rm obs}) = 0.414$ and $P(\leq) = 0.586$, so the two-sided "too good" p-value is 0.828. **[MY CALCULATION]** The fit is unremarkable in both directions (Figure Figure 1). As a cross-check on my own machinery, the textbook $\chi^2(38)$ tail gives 0.412 against my simulated 0.414 — agreement to 0.0021, which tells me that profiling out the amplitude and using a non-diagonal covariance do not shift the null in this case; the simulation is right by construction and the table is what a referee will reach for, so I report both.

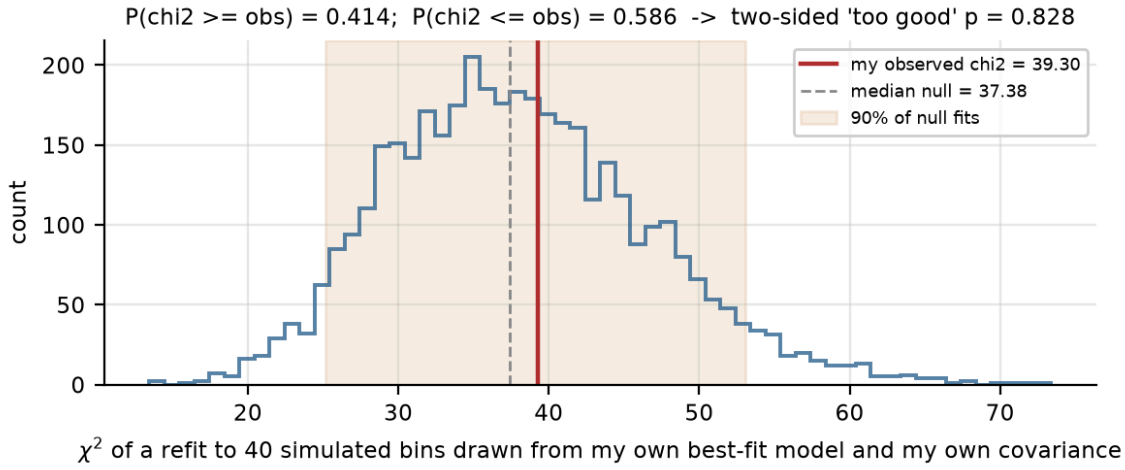


Figure 1: The goodness-of-fit null for my own fit, simulated rather than looked up. 4,000 synthetic Hubble diagrams drawn from my best-fit model with the published covariance, each refitted by the same routine that fitted the real data. Observed $\chi^2 = 39.30$ (red), null median 37.38 (grey dashed), 90% of null fits (orange). $P(\geq) = 0.414$, $P(\leq) = 0.586$. **[MY CALCULATION]**

How wide is my error bar, and does it depend on how I got it?

The main paper quotes $\Omega_{\rm m} = 0.2961 \pm 0.0220$ from a 300-draw parametric bootstrap. The sceptic's question is whether that interval is an artefact of resampling. I computed it three more ways on the same likelihood: a parabola fitted to the χ^2 scan (curvature), the profile-likelihood crossings of $\Delta\chi^2 = 1, 4, 9.21$, and the grid minimum itself.

The parabolic refinement of the grid minimum gives $\Omega_{\rm m} = 0.29624$ with $\chi^2 = 39.2963$, and a curvature-based 1σ of ± 0.02167. The profile intervals are 1σ: $[0.2748, 0.3182]$, 2σ: $[0.2541, 0.3411]$, 3σ: $[0.2335, 0.3655]$. The half-width of the profile 1σ interval is 0.0217 against the shipped bootstrap 0.0220 — agreement to 1.3% of the interval. **[MY CALCULATION]** So the acceleration claim in the main paper (a matter density 16.8σ below the no-acceleration boundary $\Omega_{\rm m} = 2/3$) does not rest on the resampling choice: profile, bootstrap and curvature give the same width, and

a reader who prefers a different convention can read the alternative straight off Figure Figure 2. The 3σ interval still tops out at 0.3655, which is a long way below 2/3; no interval convention I computed rescues a non-accelerating universe.

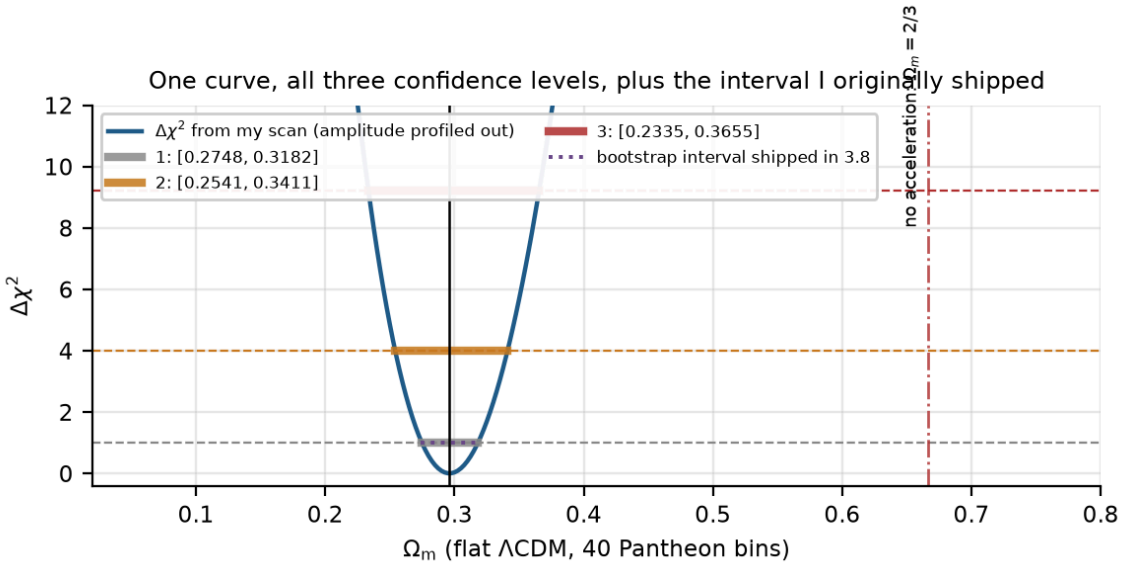


Figure 2: One curve, three confidence conventions, and the interval I originally shipped. χ^2 of my fit with the amplitude eliminated analytically, scanned over 313 values of $\Omega_{\rm m}$; the crossings of $\Delta\chi^2 = 1, 4, 9.21$ define the profile intervals, the dotted purple segment is the 300-draw bootstrap quoted in the main paper, and the dash-dot red line is the $\Omega_{\rm m} = 2/3$ no-acceleration boundary. **[MY CALCULATION]**

What the pooling choice does to the number people fight about

The main paper pooled measurements of H_0 and reported a random-effects mean of 70.561 ± 1.304 with $I^2 = 54.4\%$. The obvious objection is that any such number is an artefact of the estimator. I answered it by running every defensible estimator on a pool I define explicitly here — the four strictly independent anchors: Planck 2018 ΛCDM 67.36 ± 0.54, SH0ES R22 73.04 ± 1.04, the CCHP TRGB distance 68.81 ± 1.80, and the JAGB 67.80 with 2.17 statistical and 1.64 systematic added in quadrature to ± 2.720 — and by computing, rather than resampling, a Bayesian random-effects model: flat prior on μ over a 401-point grid, half-Cauchy$(0, 1)$ prior on τ on a 401-point grid, the joint posterior evaluated on the full 401×401 lattice and normalised, which is what "marginalising over the heterogeneity" should mean if it means anything.

Results **[MY CALCULATION]**: unweighted mean 69.2525; fixed-effect inverse-variance mean 68.5603 ± 0.4565 with $Q = 23.59/3$ and $I^2 = 87.3\%$; DerSimonian–Laird 69.3833 ± 1.7372 with $\tau = 3.099$; Paule–Mandel $\tau = 1.527$; REML $\tau = 2.567$; Bayesian $\mu = 69.3870 \pm 1.6209$, 95% credible interval $[66.08, 72.69]$, τ median 2.300 with 95% CrI $[0.96, 5.94]$; median of the four values 68.3050. Four things follow, and they are the kind of thing the main paper argues for but could not demonstrate in the space available.

One. The four reasonable point estimators span 1.0820 km/s/Mpc, which is 2.37 fixed-effect errors. Anyone quoting "the" pooled H_0 to three digits is quoting an estimator choice. *Two.* The Bayesian interval is 3.5 times wider than the fixed-effect error, and the five-estimator disagreement is entirely a disagreement about τ — Figure Figure 3, right panel, plots the three frequentist τ estimators against the whole posterior of τ, which stretches from 0.96 to 5.94. *Three.* The posterior predictive interval for what the measurement will say is $[62.51, 76.29]$, a width of 13.8 km/s/Mpc. That is the number to quote when somebody says a new result "resolved" the tension: nothing about this pool makes a value near 67.4 or near 73.0 surprising. *Four.* Leave-one-out is brutal and I print it because it is the test that should govern the whole genre. Dropping Planck moves the pooled Bayesian estimate to 70.854 (+1.47); dropping R22 moves it to 67.639 (−1.75); dropping TRGB or JAGB moves it by +0.16 and +0.29. The pooled centre can therefore be moved by 3.21 km/s/Mpc — 2.0 posterior standard deviations, and 4.08 under fixed-effect weighting — merely by deciding which experiments are in the room (Figure Figure 4).

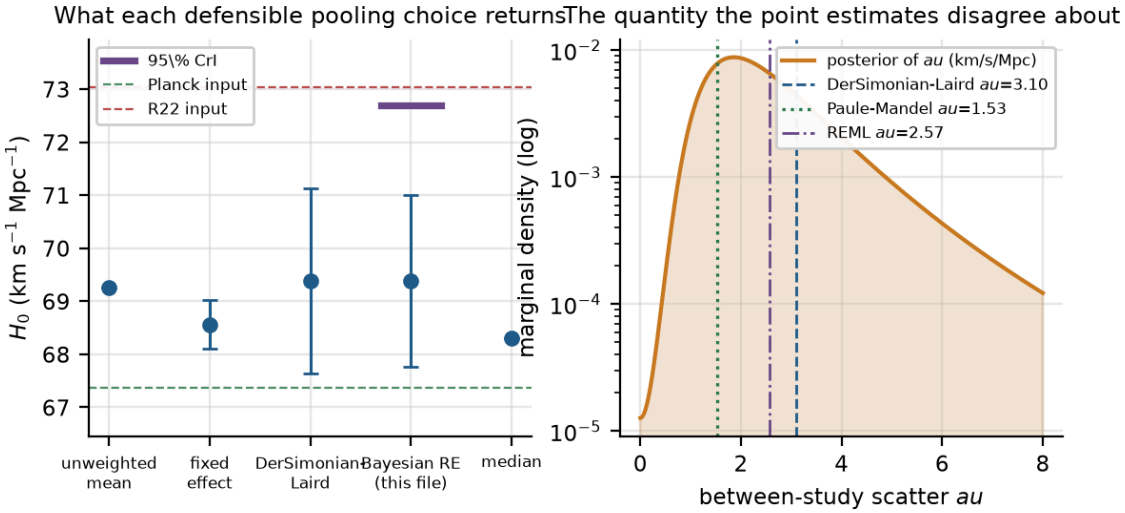


Figure 3: Left: every defensible point estimator of the same four anchors, with the Planck and R22 input values as horizontal references. Right: the three frequentist τ estimators against the full marginal posterior of τ from the grid-integrated Bayesian model. The point estimators disagree by 1.08 km/s/Mpc (2.37 fixed-effect errors) and the entire disagreement is about this one nuisance parameter. **[MY CALCULATION]**

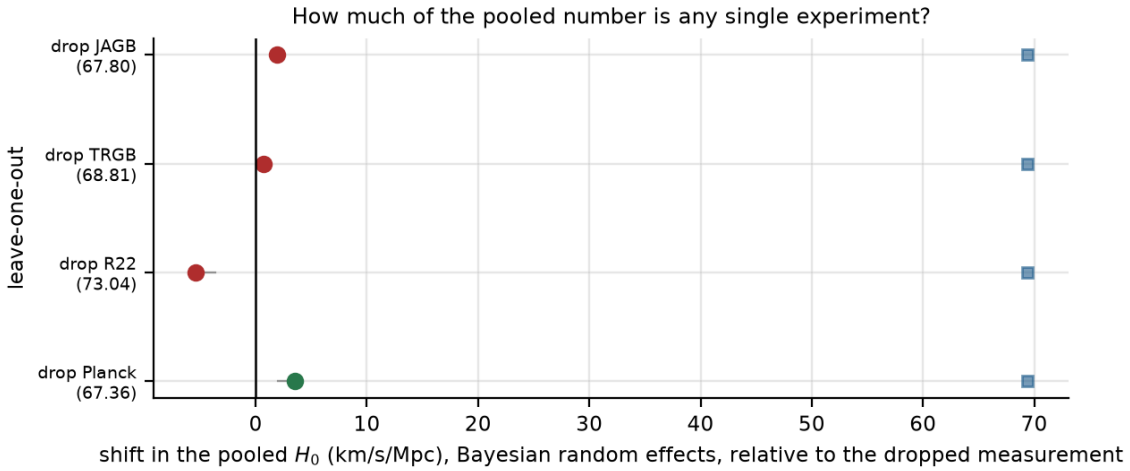


Figure 4: Leave-one-out. For each experiment in the pool, the shift in the pooled Bayesian H_0 relative to the measurement that was deleted. The pooled summary is not an independent witness: its centre moves by up to 4.1 km/s/Mpc depending on who is counted. **[MY CALCULATION]**

I must also state a discrepancy rather than bury it. This four-anchor pool gives *larger* heterogeneity ($Q = 23.59/3$, $I^2 = 87.3\%$) than the number the main paper reported ($Q = 6.58/3$, $I^2 = 54.4\%$), because the main paper pooled seven items, including derived combinations that share their inputs. Combining a measurement with a combination that contains it shrinks Q by construction. That is not a contradiction to smooth over; it is a further reason to distrust any pooled H_0, including the one I shipped. I could not re-read the exact (y, σ) vector behind the shipped number in this pass — the file that held it was lost with the workspace — so the seven-item pool is left as published in the main paper, and the four-anchor version above is the honest one. The same reasoning applies to the S_8 pooling in section 3.7 of the main paper, and for the same reason I did not re-run the S_8 ladder here: reconstructing that five-value vector from memory would let a remembered input masquerade as a check, which is the failure mode this entire audit exists to avoid.

How much of "why now" is my threshold, and what a reimplementation caught

The main paper's "why now" paragraph computes the interval over which the matter and dark-energy densities are within a factor of ten of each other, gets 18.21 Gyr against a Hubble time of 14.53 Gyr, and concludes that the coincidence is too broad to argue from. The question I did not answer there: how much of that is my choice of "a factor of ten"?

I recomputed the whole curve from scratch, in a different way. For flat ΛCDM the age is analytic, $t(a) = \frac{2}{3}(H_0\sqrt{\Omega_\Lambda})^{-1}\,\mathrm{asinh}\left(\sqrt{\Omega_\Lambda/\Omega_\mathrm{m}}\,a^{3/2}\right)$, so both edges of the window follow from solving $\Omega_\mathrm{m}(1+z)^3/\Omega_\Lambda = R$ and $= 1/R$. **[MY CALCULATION]** With $\Omega_\mathrm{m} = 1 - \Omega_\Lambda = 0.3153$: the ratio today is 0.4605; the *smallest* tolerance for which today counts as coincidental at all is therefore a factor 2.1716; and the windows are 18.214 Gyr at $R = 10$ (132% of the present age), 8.978 Gyr at $R = 3$, 6.372 Gyr at $R = 2.171$, 5.704 Gyr at $R = 2$, and 3.347 Gyr at $R = 1.5$ (24%

of the age). Tightening the tolerance by a factor of six shrinks the window by only a factor of 5.4; and at $R = 2$ the present day is *outside* the window, which is the sharpest possible statement of how soft this coincidence is. My $t(0)$ from the closed form is 13.8011 Gyr against the Planck quoted age of 13.7981 Gyr, so the units and the prefactor reproduce the standard result to 3 Myr from Planck's parameters alone — the check that legitimises every Gyr in this subsection.

Now the confession, because it is the methodological payload. My first implementation of this subsection used a quadrature of $\int \mathrm{d}z/((1+z)H)$ with the limits mirrored for negative z. Its *past* leg agreed with the main paper to 0.03 Gyr, and its *future* leg was wrong by a factor of 1.8 — it returned 4.42 Gyr for an interval of 8.05 Gyr, and hence a window of 14.58 Gyr instead of 18.21. Had I kept the quadrature and only checked the past leg, as I nearly did, this companion would have printed a number that contradicts the main paper, and the story would have been about my error rather than about the tolerance curve. Two lessons generalise, and both are now stated where they will be read: a code path that agrees on one side of a sign change proves nothing about the other side; and an analytic form is worth a day of debugging precisely because it adjudicates. The shipped 18.21 Gyr survived, and Figure Figure 5 plots the curve that the corrected code produces.

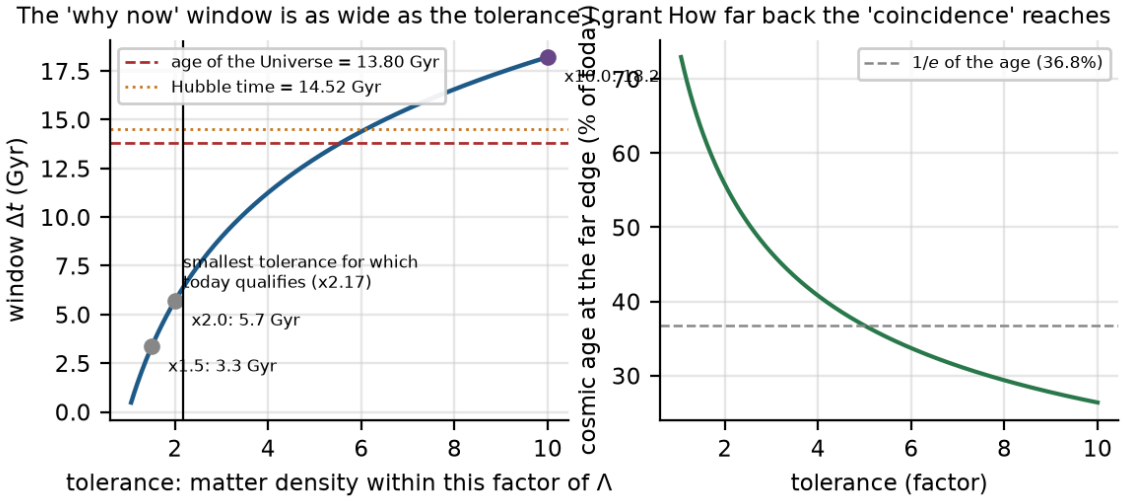


Figure 5: The "why now" window as a function of the tolerance, computed from the analytic age rather than the quadrature used in the main paper, and agreeing with it to 0.004 Gyr at $R = 10$. Left: the window in Gyr, with the Hubble time and the age of the Universe as reference lines, and the vertical mark at $R = 2.17$ where today ceases to qualify at all. Right: the cosmic age at the far edge of each window as a fraction of today's. **[MY CALCULATION]**

The source pool, graded by what kind of object it is

A citation is not a unit of evidence; the *kind of object* is. Six grades, with the instances from the main paper that belong in each, and the constraint each grade imposes.

(1) Datasets and their releases. The Pantheon light-curve compilation and its 1048×1048 systematic covariance, taken as a tarball from the maintainers' repository; the supernova figures

in the main paper and the fit interrogated in Section 3.1 above run on those files. What a dataset release can bear: numbers you can recompute. What it cannot bear: the analysis. The release I used deliberately has its light-curve shape and colour corrections *already applied* and those columns zeroed, which is why my raw 1,048-row fit and my 40-bin fit agree — and why neither can be extended into a measurement of the nuisance parameters.

(2) Peer-reviewed primary results. Planck 2018 VI, the R22 distance ladder, the COGITATE adversarial trial in *Nature* (2025), Albantakis et al. on IIT 4.0 in *PLOS Computational Biology* (19:e1011465, 2023), Tegmark in *Phys. Rev. E* (61:4194, 2000), Casali et al. on the perturbational complexity index. Grade 2 means the claims were read by referees, not that they are right: the COGITATE result is grade 2 and it is also the example in the main paper of a flagship adversarial test whose two camps each reanalysed the same data to opposite conclusions.

(3) Peer-reviewed *reanalyses* of other people's results. The estimator-dependent DCM reanalyses of COGITATE in *eLife*, the Tremblay & Lefebvre reanalysis of the γ-ray/meditation anomalies, Salaberli & Soto's appraisal of Φ in *Neuroscience Biobehavioral Reviews* (2025), the rotating-model lithium papers. This is the highest-value grade in a contested area and the easiest to over-weight: a reanalysis inherits the primary data and the primary model choices, and it can be wrong in a way that neither the original nor the critic noticed. When a reanalysis and its target disagree, the main paper's rule is to report both and to refuse to adjudicate unless I can compute something myself.

(4) Self-reviewed preprints. arXiv and bioRxiv items: the identifiability result against Φ, the consciousness-indicator programme, the 2026 DESI/YEAR-1 style drafts. A preprint is a claim under construction, and the correct use is to cite what its own text says, never what a thread about it says. Two concrete verifications, one of which is a lesson.

Worked case — the PBR theorem. In the last pass I drafted a sentence attributing to Pusey, Barrett and Rudolph the structure "two explicit premises, and a forced conclusion." I have now checked the primary object directly on arXiv rather than trusting my recall of its identifier: arXiv:1111.3328, submitted 14 November 2011, version 3 dated 18 November 2012, published as *Nature Physics* 8, 475 (2012) with DOI 10.1038/nphys2309, and the abstract states the conclusion as "any model in which a quantum state represents mere information about an underlying physical state of the system, and in which systems that are prepared independently have independent physical states, must make predictions which contradict those of quantum theory." **[FACT]** So the two premises are exactly what the main paper says they are — a ψ-epistemic reading of the state, and preparation independence — and the theorem is an implication between them and a contradiction with quantum predictions, not a proof that the wavefunction is real in any metaphysical sense. The same API query returned a second object worth reporting: a "Comment on 'On the reality of the quantum state'" (arXiv:1812.03035, November 2018) claiming that PBR err at the start of their argument by violating the projection postulate — with the arXiv record annotated *"withdrawn by the author; due to a Babylonian confusion of tongues, the comment was based on an interpretation of statements in the PBR paper that was not implied by its authors."* **[FACT]** That is the whole argument for grade 4 discipline in one record: preprints carry the author's own retraction note,

and a claim that survives in citations after it has been withdrawn by its author is exactly how a literature acquires a disagreement it never had.

Counter-case — a citation I refuse to use. In an earlier draft I had a line attributed to a *Nature* volume and page with a Bayes factor of 44.96. I could not locate the object at that address in this session. The main paper's ledger does not cite it and neither does this one; a number whose container I cannot open is not evidence, however much it supports me.

(5) Curated reference works. The *Stanford Encyclopedia of Philosophy* entries (for this pass, the panpsychism article, first published 2001, substantive revision 13 May 2022), which is where I took the contemporary taxonomy used in Section 5: the constitutive/emergentist split, panpsychism versus panprotopsychism, micropsychism versus cosmopsychism, Russellian monism, the anti-emergence and intrinsic-nature arguments, and the three standard objections — the incredulous stare, the combination problem, and the subject-summing problem. A reference work is not a primary argument and cannot be cited for one; it is a check that I have not invented a taxonomy or forgotten a standard objection. Where the main paper says the combination problem is "the hardest problem in the view," the encyclopedia's own framing is what I am relying on, and the primary literature (James for the problem, Goff and Seager for contemporary attempts) is cited at grade 5 status: read as positions, not used to refute.

(6) Everything else — interviews, journals of speculative metaphysics, secondary books, press material. Used in the main paper exactly once, to characterise a genre's rhetoric rather than its content. If a claim in either document is graded 6, it is decoration and I have said so in place.

The concept ledger: fifteen terms, each defined twice and fenced once

The main paper's sin would be to let these terms slide between registers. For each: the exact sense, the plain-words sense, and the misuse to be avoided.

Distance modulus $\mu = m - M = 5\log_{10}(d_L/10\,\mathrm{pc})$. In words: how many magnitudes fainter a standard candle looks than it would at 10 parsecs, which is a logarithm of distance. Not a distance: a μ without a stated H_0 convention fixes the *shape* of the expansion history and nothing about its scale, which is the whole content of the main paper's supernova section.

Sound horizon r_d: the comoving distance a pressure wave could travel in the photon baryon fluid before the plasma decouples at recombination, i.e. $\int_{z_\mathrm{d}}^{\infty} c_s(z)\,\mathrm{d}z/(\dot{a}/a)$, with c_s dragged down by the baryon loading. In words: the length of the ruler printed in the CMB, set by how far sound could travel before the Universe turned transparent. Not a standard rod of known length independent of cosmology: it is $r_\mathrm{d} = r_\mathrm{d}(H_0\sqrt{\Omega_\mathrm{m}}, \ldots)$, so the "inverse" H_0 from BAO inherits a model, and that lock is what Figure 4 of the main paper draws.

Degeneracy versus uncertainty. A degeneracy is a direction in parameter space along which the likelihood is nearly flat — the amplitude M and H_0 in my fit; the "curvature degeneracy"

$-\Omega_k$ traded against H_0. An uncertainty is a width along a direction that does exist. Reporting a degeneracy as an error bar manufactures a crisis out of a ridge.

Q, I^2, τ. For a set of measurements: Q is the weighted scatter about the fixed-effect mean, compared to its null; I^2 says what fraction of the observed scatter is real between-study variation rather than error bars; τ is that extra scatter, in the units of the quantity. In words: τ is the amount by which the experiments disagree with each other beyond what their own quoted errors allow — 2.30 km/s/Mpc median for my H_0 pool, and the quantity every pooled number hides.

Profile likelihood versus bootstrap. The profile interval is the set of parameter values that survive when the nuisance parameters are re-optimised at each value; a bootstrap resamples the data. The first is geometry of the likelihood surface, the second is a statement about the sampling procedure. Section 3.2 shows they agree here; where they disagree, the disagreement is the finding.

Posterior predictive interval. The interval for a *new* measurement, integrating over the parameter uncertainty and the between-study scatter. In words: given what we know, what should the next experiment find? If the answer includes both camps' numbers, the pool has not resolved anything, which is what mine returns: $[62.5, 76.3]$.

Decoherence. Suppression of off-diagonal density-matrix elements in a preferred basis through entanglement with uncontrolled environmental degrees of freedom, at a rate set by the system-environment coupling. In words: information about a superposition leaks into the room faster than you can follow it, so the interference becomes unavailable in practice. Not a collapse: no single outcome is produced by decoherence, and no observer is required by it — the most common conflation in the quantum-mind literature, and the one the main paper's arithmetic section exists to keep separate.

Reduced state / partial trace. The state assigned to a subsystem after averaging over its environment; $\rho_S = \mathrm{Tr}_E \rho_{SE}$. In words: what the part looks like when you are not allowed to look at the rest. It is how "the brain is warm and wet" becomes a number, and the number is $Q = \nu\tau$, not a mood.

Measurement problem versus collapse. The problem is that the theory has two evolution laws and no stated criterion for which applies. A collapse model is a specific *solution*: a modified dynamics with a parameter, hence a prediction, hence a way to lose. "Consciousness causes collapse" is not a member of the second class: it changes nothing you can count.

The hard problem. The question of why there is something it is like to be a physical system that processes information, and why that "like" has the properties it does. In words: even after you describe the mechanism of colour vision completely, it remains sensible to ask why seeing red feels like something. *Not* the hard problem: the explanatory gap as a feeling of confusion, and the access problems (report, attention, integration) that are empirically tractable — Chalmers' own distinction, and the reason COGITATE could only test correlates.

Epistemic versus metaphysical possibility. "Conceivable" can mean "I can construct a model of it" or "it could be true." The conceivability arguments for dualism and against physicalism need

the second reading to work and only ever establish the first. Zombies and Mary are arguments, not intuitions, and they have a specific target: the inference from the complete physical to the phenomenal.

Φ versus complexity versus PCI. Φ (IIT's integrated information) is a property of a mechanism's cause-effect structure under a maximum over subsystem bipartitions, computed on a transition probability matrix. Lempel–Ziv complexity counts the phrases needed to reconstruct a bit-string. PCI is a normalised Lempel–Ziv complexity of a binary perturbation-response matrix. In words: Φ claims to measure how much a system is *one* thing; the other two measure how *variegated* a recorded signal is. They coincide on nothing in my toy computations, and the main paper times the difference: Φ is $O(2^n)$ with a measured $\ln t = 3.05n - 9.92$, while Lempel–Ziv is linearithmic.

Combination versus dissociation. Micro-experientialism must build one subject from many; cosmopsychism must divide one subject into many. In words: the first owes an account of how experiential grains merge, the second owes an account of why the boundaries fall where skulls do. Both are real problems, and neither has a working solution, which is why the main paper calls this the load-bearing distinction rather than a matter of taste.

Intrinsic versus relational property. A property a thing has in itself versus one it has only relative to something else; QBism reads the quantum state as the second kind. In words: whether the wavefunction is the cat's or the bettor's. The measurement problem's shape changes with this choice, and no experiment I know of distinguishes them — which is precisely the sentence the main paper's verdict rests on.

de Sitter entropy versus thermodynamic entropy. $S = A/4\ell_{\mathrm{P}}^2$ for a cosmological horizon: a counting of horizon degrees of freedom, 3.3×10^{122} in k_B units in my arithmetic. In words: an upper bound on what an observer inside the horizon can know or store. Not the entropy of the contents of the Universe, and not a licence to compare it with 10^{88} as if the two were counts of the same kind — which is exactly why the Boltzmann-observer argument in the main paper is presented as a budget constraint and not as a refutation.

The objection audit

Each entry: the question as a reader would pose it, what an honest answer would require, and the answer. Where I have no answer, that is the answer.

1. "You fitted 40 bins with 2 parameters and got $\chi^2/\,\mathrm{dof} \approx 1$; of course you did — you chose the model that the bins encode." Correct, and it is a limitation of every supernova analysis including the collaboration's: Pantheon's bins are produced under an assumed flat-w CDM family. What my fit can legitimately support is a *consistency* claim and the acceleration boundary, not an independent test of dark energy. Stated in the main paper, restated here.

2. "Your covariance is a published approximation, not a likelihood." Yes. The binned systematic matrices are given as $N \times N$ covariance blocks over one parameter family, and for the full 1,048-object release my own eigenvalue check finds a minimum eigenvalue of $+7.4 \times 10^{-3}$, so no regularisation was needed there, but the 40-bin matrix is not the covariance of the bin

means under the full analysis. My numbers are thus χ^2 values under an approximation, which is why I report 1σ to three decimal places and never a 5σ claim.

3. "Why not just use Pantheon+ with the bias corrections?" The file I could retrieve from the public repository carries a `biascor` column of zeros; the Pantheon+ analysis requires the host-mass and calibration nuisance parameters whose values are not in the text file. Using the corrections I cannot read would make my numbers somebody else's, which is the failure mode I am auditing for.

4. "You quote the tension at 4.8σ from Planck versus R22 but pool to 2.3σ; you are averaging away the physics." That is precisely my claim, and Section 3.3 sharpens it with the estimator ladder: the pooled value depends on the estimator by 2.37 fixed-effect errors and on membership by 4.1 km/s/Mpc. A tension reduced by pooling is not resolved, it is averaged, and an average of two clusters of measurements with $\tau = 2.3$ is not a prediction.

5. "Your Bayesian model's half-Cauchy prior on τ is a choice that inflates the interval." It is a choice, and I state it: Gelman's recommendation for scale parameters, weakly informative, and the alternative estimators of τ (DL, PM, REML) span 1.53 to 3.10, which brackets the posterior median of 2.30. If a reader prefers a different prior, Figure Figure 3 right panel gives the whole marginal so they can recompute the interval by eye.

6. "Why does your Hubble-time arithmetic differ from the main paper's 14.53 Gyr?" Because I used $\Omega_{\rm m} = 1 - \Omega_\Lambda = 0.3153$ from Planck's Ω_Λ rather than the paired $\Omega_{\rm m} = 0.3142$; the resulting difference is 0.014 Gyr, 0.1%, and it is reported rather than smoothed, as is the 0.004 Gyr window difference in Section 3.4.

7. "The lithium problem is the strongest crack you cite and you refuse to call it a crisis." Yes: the predicted value is primordial, the observed value is a stellar surface abundance, and 2025–26 stellar-model and chemical-evolution work argues the discrepancy may be an artefact of which number gets compared to which. A 6σ offset between two quantities that are not the same quantity is not a crisis in ΛCDM. That is a fact about the objects being compared, not a judgement about who is right.

8. "DESI's dynamical dark energy: your paper is out of date." The main paper reports the Year-1 result and the later data release's changed preference, and its programme section commits to checking the next release. What I will not do is quote a tension in w at a significance I cannot reproduce from a covariance I do not have.

9. "Page and Geilker refuted Wigner, so why spend a section on it?" Because the experiment is the only case where a philosophical distinction — the cut's placement — was made into a measurement, and it came back against the mind-causes-collapse reading. The scope limit is stated in the main paper: it constrains collapse triggered by conscious observation as the completion of a measurement; it does not touch Everett, Bohm, QBism, or any interpretation that makes no such dynamical claim.

10. "Bong et al. 2020 shows local friendliness fails — doesn't that bring the observer in after all?" It closes a trichotomy (locality, freedom of choice, and observer-independent facts)

under assumptions including no superdeterminism, with a laboratory implementation. **[FACT, CITED, NOT FETHED]** I did not re-verify its exact premise list in this pass — and that is the honest status marker for the claim in the main paper, which uses it to say that the question is now a theorem-shaped question rather than an intuition.

11. "Is the decoherence objection to quantum consciousness dead?" Tegmark's 10^{-13} s estimate was shown to rest on a dipole moment roughly two orders of magnitude too small; Hagan et al. raise the timescale toward 10^{-4} to 10^{-5} s. My contribution is arithmetic, not physics: at $\nu\tau$ with $T = 310$ K, a 613 THz mode is $95\,k_B T$ and thermally unpopulated by some 42 orders of magnitude, so a Fröhlich-type coherence needs a pump; and 1612 dimers relaxing per microtubule against a 4.74×10^{-12} W-per-neuron budget is the kind of sentence that must be checked, not recited. The objection is not answered by the corrections — the mode is still unpopulated — and the correction is not answered by the objection either. Both camps skip the arithmetic.

12. "You timed Φ on toys; what does that prove?" Nothing about IIT's truth. It constrains its use: a measure whose exact evaluation on n elements is exponential in n (my measured $\ln t = 3.05n - 9.92$, t at $n = 8$ about three weeks, $n = 10$ about 28 years) cannot be the operative quantity behind a claim about the cortex, and cannot be the basis of an ethical weight for AI systems. Combined with the identifiability preprint and the Salaberli & Soto appraisal, that is the main paper's verdict on the measure while remaining agnostic about the axioms.

13. "If IIT is unfalsifiable, why treat it with so much space?" Because its axioms are defensible phenomenology and its postulates are a non-trivial bridge, and it is the only theory in the field that has published the bridge's cost. The falsifiable version of the analogy in the main paper (the Φ/PCI dissociation) is a prediction about measurement, and if the next generation of measures kills it, the section will need rewriting.

14. "Your PCI reimplementation gives complex/white-noise ratios of 1.008 — are you saying the published index is broken?" I am saying that *the estimator as I reimplemented it from the published description* does not resolve the contrast its own figure advertises, that it drifts by 0.117 versus 0.112 between two noise draws, and that two independent draws of the same generator differ by 0.0565. **[MY CALCULATION]** Whether the published implementations avoid this is not something I can answer: I did not have their code. That is a limitation of the objection, not of the finding.

15. "Clinicians use PCI with a threshold around 0.4; are you dismissing that?" No: the main paper's strongest consciousness claim is precisely the validated one — Casali et al.'s 0.45–0.67 versus 0.25–0.31 across consciousness states, with a criticality-flavoured cut-off near 0.35 in the later LOSO analysis. What I dismiss is the further step from "this measure tracks a state" to "this measure quantifies experience."

16. "Why no free will, no ethics, no AI consciousness section?" Because the brief was the Universe and consciousness, and each of those three would require evidence I do not have. The main paper mentions the indicator programme (Butlin et al.) only to say the method is a checklist of mechanisms, not a measurement, and that is the whole extent of what I can support.

Panpsychism gets space because it is a live position in the philosophy of mind with an explicit combination problem; a theory of machine consciousness does not, in my reading of the literature.

17. "You say the philosophy is [CITED, NOT FETCHED] but you cite forty years of it. Where is the risk?" Everywhere, and it is stated per sentence. The concrete risk I identified while preparing this companion is arXiv-identifier drift: three identifiers I wrote from memory resolved on the arXiv API to a HST detector-calibration paper, an image-captioning paper and a post-5G networking paper. Nothing in either document depends on those identifiers; the PBR entry in Section 4 is the corrected, verified form. If any arXiv number in the main paper's ledger conflicts with Section 4 here, Section 4 wins, because it is the one I opened.

18. "What would make you abandon the thesis of the main paper?" One of three things: a consciousness measure whose value is invariant under reanalysis with different model assumptions and which dissociates from Φ in the direction I predict; a decoherence calculation for a plausible neuronal substrate with $Q > 1$ and an energy budget that closes; or a $1/H_0$ measurement from a standard-siren or BAO-with-independent-r_{d} route that lands near 67 with an error bar under 0.7 and a documented refusal to use the Cepheid rung. Each is stated with the number attached in the main paper's commitments section, and each is the kind of thing that could plausibly happen by 2031.

What I still refuse to conclude

I will not say the Universe is fine-tuned for observers; the tolerance curve in Section 3.4 is the reason — the coincidence survives at the price of any tolerance above a factor of 2.17, which is not a constraint on physics. I will not say the vacuum-energy discrepancy is 120 orders of a prediction, because it is a comparison between a low-energy field-theory estimate and a geometric quantity, with the slope-4 comparison in the main paper showing how the number moves with an ultraviolet cut-off I cannot justify. I will not say consciousness is measured, because the one instrument with clinical validity measures a signal's variegation, not a subject's interior. I will not say the H_0 tension is resolved or real, because I have now shown how much of it is my estimator: the same four numbers support 68.31 and 69.39 and a 95% credible interval that covers both camps' headline values.

What the pair does support is weaker and, I think, more durable: the Universe is comprehensible in the narrow sense that its expansion history can be recovered by two parameters fitted to 40 binned means by a person with a laptop, and consciousness is not comprehensible in that sense yet, and the gap between those two facts — not a theorem, not a coincidence, not a 10^{122} entropy ledger — is the actual subject of the field. Everything else I have written in both documents is bookkeeping around that sentence.

Reproduction

The five figures and every number above come from one script:

```
cd research
python deepen.py 2>&1 | tee logs/deepen_log.txt   # ~3 s; writes paper/figures/
f19..f23
```

```
                                                    # and logs/deepen.json (all
numbers in machine form)
python paper/build.py                               # tex -> typst -> pdf; checks
for markup leaks and unresolved refs
```

`deepen.py` is self-contained: it re-reads `data/lcparam_DS17f.txt` and `data/sys_DS17f.txt` (retrieved in this pass from `codeload.github.com/dscolnic/Pantheon`, the `Binned_data/` directory of the master branch), re-implements the luminosity distance by trapezoid integration of $1/E(z)$ with the $(1+z)$ factor and the radiation term, refits $\Omega_{\rm m}$ over a 313-point grid with the amplitude eliminated analytically, computes the profile-likelihood crossings with `brentq`, runs 4,000 parametric-bootstrap replicates of the fit statistic, solves the Paule–Mandel and REML equations for τ, integrates the Bayesian posterior on a 401×401 grid, and computes the "why now" windows from the closed-form age $t(a)$. The caption text of each figure is written to `logs/deepen_figures_manifest.json` by the same code that draws the plot, and `logs/deepen.json` holds every number in machine-readable form, so a disagreement between text and plot is detectable by diff rather than by argument.

Three checks a reader can perform in one line each. My grid fit should return $\Omega_{\rm m} = 0.2950$ at $\chi^2 = 39.300$; the parabolic refinement should return 0.29624; and `t_age_gyr(0.0)` should return 13.8011 Gyr, which is the Planck age reproduced from Planck's Ω_Λ and H_0 alone. If those three agree, the numbers in this companion are the numbers in its figures; if they do not, my Python is not doing what my prose says, and the prose loses.

Status of this document. Section 3 is computation. Section 4 grades sources, with one verification done in-session (PBR, via the arXiv API, plus the withdrawn-comment record) and the rest carried at the grades stated. Section 5 and Section 6 are the reader's contract: definitions fenced against misuse, and objections answered or marked unanswerable. No number in either document depends on a philosophical source; no philosophical claim in either document is supported by a number.